Problem Set X, Macroeconomics III
University of Copenhagen

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Exercise 1 – Price setting with Fisher contracts

Consider the following model of price setting under imperfect competition, similar to the one studied in lectures. Firms predetermine prices for two periods as a function of demand and the prices (fixed or expected) of their competitors. We assume symmetry such that half the firms make decisions in even periods and the other half in odd periods. Firms can set different prices across periods. The notation is as follows: p_t^2 is the price that was set in period $t-2$, for period t.

The following equations characterize the economy:

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\begin{aligned}
p_t^i &= \phi \mathbb{E}_{t-i}\left[m_t\right] + (1-\phi)\mathbb{E}_{t-i}\left[p_t\right], \quad \phi \in (0,1) \\
p_t &= \frac{1}{2}\left(p_t^1 + p_t^2\right) \\
y_t &= m_t - p_t.
\end{aligned}
$$

The first equation describes how prices are set as function of expected demand and competitors prices, the second equation describes the price level, the final equation provides us with aggregate demand. Expectations $\mathbb{E}_t[\cdot] \equiv \mathbb{E}\left[\cdot \mid I_t\right]$ are based on information available in at time t, summarized in I_t. This information includes all current and past values of m_t.

a) Find $\mathbb{E}_{t-2}\left[p_t^1\right]$ as a function of m_t and expectations of m_t.

b) Similarly, find expressions for p_t^1 and p_t^2.

c) Find the equilibrium values for p_t and y_t as functions of m_t and expectations of m_t.

d) Do monetary policy shocks m_t have persistent effects on output. Discuss.

Now, to see the role that monetary policy can play in stabilizing output fluctuations, assume that nominal demand can be decomposed into:

$$y_t = m_t - p_t = m_t^{cb} + v_t - p_t$$
$$v_t = v_{t-1} + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}\left(0, \sigma_\epsilon^2\right),$$

where

$$m_t^{cb} = a_1\epsilon_{t-1} + a_2\epsilon_{t-2}. \quad a_j \in \mathbb{R}.$$

One can think of m_t^{cb} as being the part of nominal demand controlled by the central bank, and v_t being the "uncontrollable" part (i.e., stochastic money velocity).

e) Find the values of a_1 and a_2 that minimizes output volatility. Explain the intuition behind the result.[1]

f) What are the values of a_1 and a_2 that instead minimize price volatility? Is there a conflict between the goals of price and output stabilization? Is there "divine coincidence"?

[1] You can replace m_t by $m_t^{cb} + v_t$ in the previous solution.